

1. Local hidden variable models I

Alice and Bob share a two-qubit quantum state and each perform one of two measurements on their state, labelled by x and y , with outcomes a and b respectively. x and y take values 0 and 1, while a and b take values 1 and -1 .

(a) Assume first their measurements lead to the following probabilities

$$P(a, b|x, y) = \begin{cases} \frac{1}{2} & \text{if } a = b, x = 0 \text{ (for any value of } y), \\ \frac{1}{4} & \text{if } x = 1 \text{ (for any value of } a, b, y), \\ 0 & \text{if } a \neq b, x = 0 \text{ (for any value of } y). \end{cases}$$

Find a local hidden variable model which is able to reproduce these probabilities.

(b) Assume now their measurements lead to the following probabilities

$$P(a, b|x, y) = \begin{cases} \frac{2}{3} & \text{if } a = b = 1, x = y, \\ \frac{1}{3} & \text{if } a = b = -1, x = y, \\ \frac{1}{3} & \text{if } (a, b) \neq (-1, -1), x \neq y, \\ 0 & \text{otherwise.} \end{cases}$$

Find a local hidden variable model which is able to reproduce these probabilities.

2. Local hidden variable models II

Alice, Bob and Charlie share the joint state of three qubits

$$|\Psi\rangle = \frac{1}{2} (|0\rangle|0\rangle|0\rangle - |0\rangle|0\rangle|1\rangle + |1\rangle|0\rangle|0\rangle - |1\rangle|0\rangle|1\rangle)$$

- (a) Calculate the reduced density operators ρ_A, ρ_B and ρ_C for each of the three qubits.
- (b) Using part (a) show that $|\Psi\rangle$ is a product state, and write it explicitly in product form.
- (c) Suppose that Alice, Bob and Charlie make either X or Z measurements on their qubits. Since $|\Psi\rangle$ is a product state, the probabilities can be factorized as

$$P(a, b, c|x, y, z) = P_A(a|x)P_B(b|y)P_C(c|z)$$

Find the probabilities $P_A(a|x)$, $P_B(b|y)$ and $P_C(c|z)$ in this case.

- (d) Give a local hidden variable model which can reproduce the results of this experiment, by specifying a probability distribution $P_S(\lambda, \mu, \nu)$ for hidden variables $\lambda = (a_0, a_1)$, $\mu = (b_0, b_1)$ and $\nu = (c_0, c_1)$ which determine Alice, Bob and Charlie's measurement outcomes.

Hint: Because the probabilities factorize into a product of three distributions, the model will also factorize into three local strategies, one for each party (which should help in finding it).

3. The CHSH game

The bound proved by Clauser, Horne, Shimony and Holt for local hidden variable theories is often presented as the inequality

$$E(A_0B_0) + E(A_0B_1) + E(A_1B_0) - E(A_1B_1) \leq 2 \quad (6.1)$$

where A_0 and A_1 indicate the different measurements by Alice, with outcomes $+1$ or -1 , B_0 and B_1 indicate the different measurements by Bob, with outcomes $+1$ or -1 , and $E(A_xB_y)$ corresponds to the expectation value of the product of Alice and Bob's outcomes when Alice measures A_x and Bob measures B_y on their respective systems.

- (a) Let Alice's and Bob's outputs be denoted by α and β respectively, and define new variables $a, b \in \{0, 1\}$ such that

$$\alpha = (-1)^a \quad \beta = (-1)^b.$$

Show that

$$E(A_x B_y) = 2 \text{Prob}(a = b | x, y) - 1$$

where $\text{Prob}(a = b | x, y)$ is the probability that $a = b$ when Alice measures A_x and Bob measures B_y .

- (b) Show that

$$a \oplus b = xy \implies \begin{cases} a \neq b & \text{if } x = y = 1 \\ a = b & \text{otherwise} \end{cases}$$

where $\oplus$ denotes addition modulo 2.

- (c) If x and y are chosen uniformly at random, i.e. $\text{Prob}(x, y) = \frac{1}{4}$ for $x, y \in \{0, 1\}$, show that equation (6.1) is equivalent to the relation

$$\text{Prob}(a \oplus b = xy) \leq \frac{3}{4}. \quad (6.2)$$

- (d) It is also true that for any local hidden variable model

$$-2 \leq E(A_0 B_0) + E(A_0 B_1) + E(A_1 B_0) - E(A_1 B_1)$$

Find a bound on $\text{Prob}(a \oplus b = xy)$ which is equivalent to this new inequality. Can you give an intuitive explanation for why this new bound must hold if (6.2) does.

4. Quantum nonlocality

Consider a game G in which Alice and Bob each get given a random bit, x and y respectively, such that the four possibilities

$$x = 0, y = 0, \quad x = 0, y = 1, \quad x = 1, y = 0, \quad x = 1, y = 1,$$

each occur with probability $1/4$. Alice and Bob must produce output bits a and b respectively. In order to jointly win the game G their outputs must satisfy

$$a \oplus b = xy \oplus x \oplus 1 \quad (6.3)$$

where $\oplus$ denotes addition modulo 2. Alice and Bob are allowed to agree a strategy in advance, but cannot communicate during the game.

- (a) A *deterministic* local strategy corresponds to specifying a_0, a_1, b_0 and b_1 . Write down four equations which must be satisfied by these variables in order for the strategy to win the game G in all cases. Using these equations, show that no deterministic local strategy can win G with certainty.
- (b) Give an example of an optimal deterministic local strategy for G , and specify its probability of winning the game. (Note that a probabilistic local strategy cannot do better than the best deterministic local strategy)
- (c) Alice and Bob can design a quantum procedure, in which they each measure a local operator on an entangled quantum state, such that the probabilities for their outcomes a and b , given their input bits x and y , satisfy

$$P(a = b | x, y) = \frac{1}{2} (1 + \cos(\theta_x - \phi_y))$$

$$P(a \neq b | x, y) = \frac{1}{2} (1 - \cos(\theta_x - \phi_y))$$

where $\theta_0, \theta_1, \phi_0$ and ϕ_1 are angles in the interval $(0, 2\pi)$, such that $\theta_x = \theta_0$ if $x = 0$ etc. Choose values for these four angles such that Alice and Bob win the game G with higher probability using their quantum strategy than the optimal local strategy considered above.